Adiabatic Gravitational Waveforms (Weak-field) in Schwarzschild - de Sitter

In this notebook we show the plots of gravitational waveforms for compact objects around Schwarzschild black holes influenced by an SdS parameter λ (y in this notebook). Results of this notebook has been used in Villanueva, John Adrian N., and Ian Vega. “Extreme-mass ratio inspirals in Schwarzschild-de Sitter spacetime: Weak-field orbits.” . https://journals.aps.org/prd/pdf/10.1103/npk4-61t8

The waveform polarizations are obtained using numerical kludge by

$$h_+ = \frac{1}{D}\left(\ddot{Q}_{11} - \ddot{Q}_{22} + 2\sqrt{\frac{\Lambda}{3}}(\dot{Q}_{11} - \dot{Q}_{22})\right) + \frac{\Lambda}{3}\left(\dot{Q}_{11} - \dot{Q}_{22} + \frac{2\sqrt{\Lambda}}{3}(Q_{11} - Q_{22})\right), \quad (4.19)$$

$$h_\times = \frac{2}{D}\left(\ddot{Q}_{12} - 2\sqrt{\frac{\Lambda}{3}}\dot{Q}_{12}\right) + \frac{\Lambda}{3}\left(\dot{Q}_{12} - 2\sqrt{\frac{\Lambda}{3}}Q_{12}\right). \quad (4.20)$$

With the quadrupolar moment as

$$Q_{ij} = \mu r^2(\psi)\begin{pmatrix} \cos^2(\phi) & \sin(\phi)\cos(\phi) & 0 \\ \sin(\phi)\cos(\phi) & \sin^2(\phi) & 0 \\ 0 & 0 & 0 \end{pmatrix}. \quad (4.21)$$

The derivatives in Q translate to derivatives in $r(\psi)$ which uses the solutions of the weak field orbital evolution, included in this repository

Preamble

In[ ]:=
```
Exit[]
```

```
ClearAll["Global"]
```

In[24]:=
```
plotsFolder = FileNameJoin[{ParentDirectory[NotebookDirectory[]], "plots"}];
(*automatically saves to the plots folder in the repo*)
```

In[16]:=
```
Get["MaTeX`"];(*MaTeX for plot generation*)
SetOptions[MaTeX, FontSize → 19,
  "Preamble" → {"\\usepackage{newtxtext,newtxmath}"}, Magnification → 1.2];
style = Directive[FontFamily → "Times", FontSize → 23, Black];
```

Orbital trajectories from the WFSDSORB package

In[2]:=
```
SetDirectory[NotebookDirectory[]];
```

Pulling up the package

In[3]:=
```
<< "../source/WFSDSOrbits.wl"
```

Main orbit solver and plotter

In[4]:=
```
?wfsdsorb
```

Out[4]=

Symbol

wfsdsorb[p0,e0,y0,tf0] plots the SdS orbit (for a

given y0) vs a reference 2.5PN y=0 orbit, with fixed (p0, e0) up to time tf0

In[5]:=
```
?sol
```

Out[5]=

Symbol

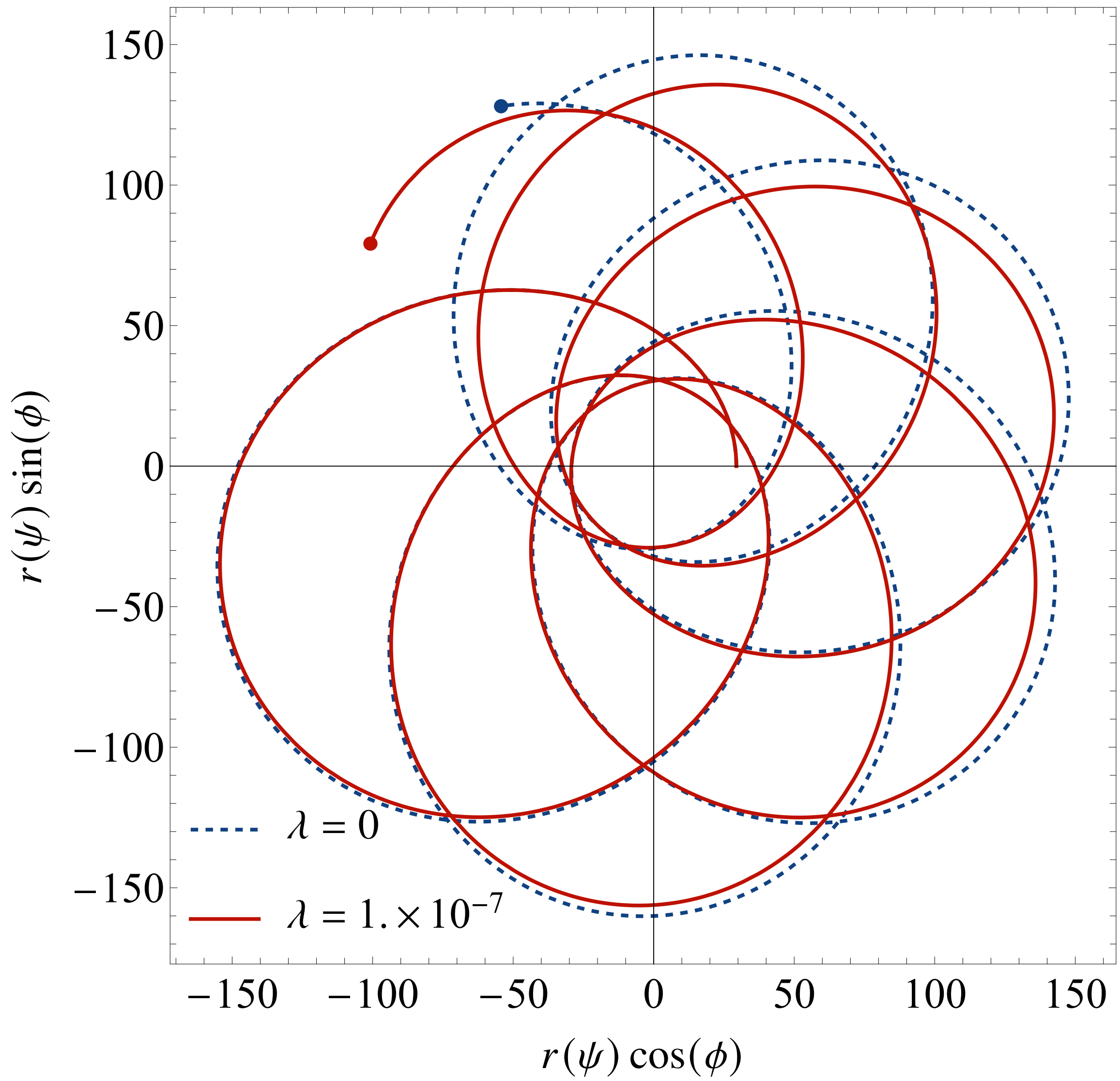
sol[p0,e0,y] computes the evolution of osculating elements (p,e,f,δ) for initial p0 and e0, for a given SdS parameter y

Sample orbit, p=50, e=0.7, $\lambda = 10^{-7}$ up to t=30000

In[6]:=

```
wfsdsorb[50, 0.7, 10^-7, 30000]
```

Out[6]=



One could also explicitly compute the coordinates of the EMRI in x-y plane

In[7]:=

```
{?xpos, ?ypos}
```

Out[7]= { Symbol: xpos computes x position of emri at time t , Symbol: ypos computes y position of emri at time t }

In[10]:=

```
{xpos[t], ypos[t]}
```

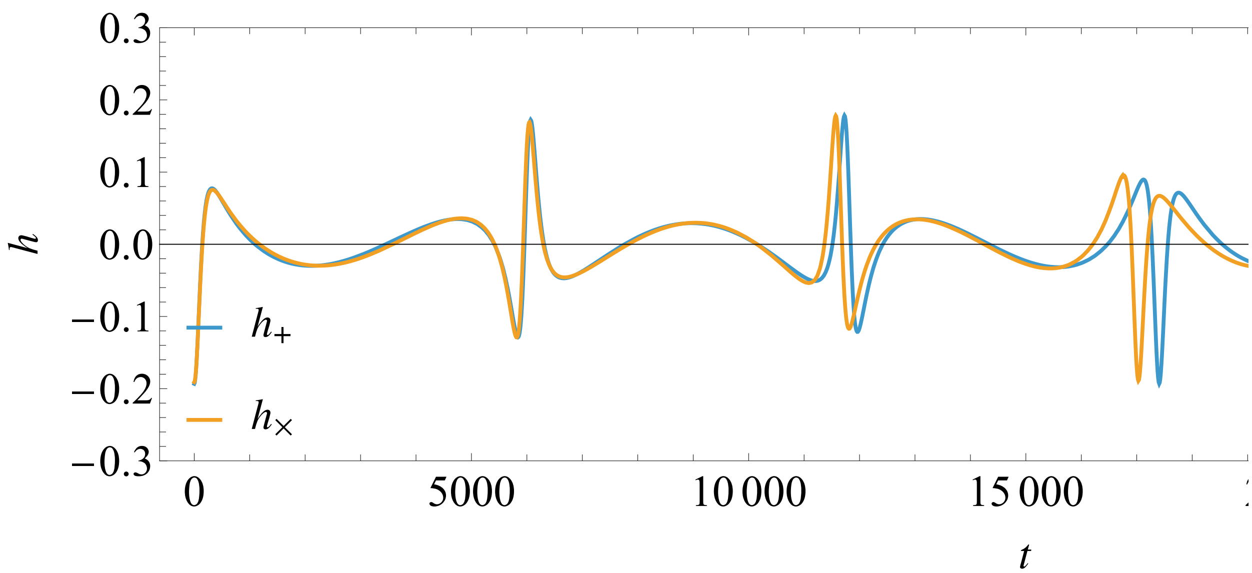
Out[10]=

$$\left\{\frac{\text{Cos}[f[t]+\delta[t]]\,p[t]}{1+\text{Cos}[f[t]]\,e[t]},\ \frac{p[t]\,\text{Sin}[f[t]+\delta[t]]}{1+\text{Cos}[f[t]]\,e[t]}\right\}$$

One could then build the polarizations from the coordinates, e.g. $h_{+} \sim \ddot{Q}_{11} - \ddot{Q}_{22} \sim \ddot{x}^2 - \ddot{y}^2$ with radiation reaction substituted from the orbital evolution solvers

In[19]:=
```
Plot[{Evaluate[D[xpos[x]^2 - ypos[x]^2, {x, 2}] /. sol[50, 0.7, 0]],
  Evaluate[D[xpos[x]^2 - ypos[x]^2, {x, 2}] + 2 Sqrt[10^-7]/3 D[xpos[x]^2 - ypos[x]^2, {x, 1}] /.
    sol[50, 0.7, 10^-7]]}, {x, 0, 30000}, ImageSize → 1000, Frame → True,
 AspectRatio → 1/4, PlotRange → {Automatic, {-0.3, 0.3}}, LabelStyle → style,
 Frame → True, FrameLabel → {MaTeX["t"], MaTeX["h"]},
 PlotLegends → Placed[{MaTeX["h_+"], MaTeX["h_\\times"]}, {Left, Bottom}]]
```

Out[19]=



We can then build functions for the polarizations

```
hplus[x0_] := Evaluate[D[xpos[t]^2 - ypos[t]^2, {t, 2}]] /. t → x0
hcross[x0_] := Evaluate[D[2 xpos[t] × ypos[t], {t, 2}]] /. t → x0
hplusSdS[y_, x0_] :=
 Evaluate[D[xpos[t]^2 - ypos[t]^2, {t, 2}]] + 2 Sqrt[y] D[xpos[t]^2 - ypos[t]^2, {t, 1}] /. t → x0
hcrossSdS[y_, x0_] :=
 Evaluate[D[2 xpos[t] × ypos[t], {t, 2}]] + 2 Sqrt[y]/3 D[2 xpos[t] × ypos[t], {t, 1}] /. t → x0
```

We can also build a function for the polarizations for a given orbital parameter

In[11]:=
```
hplusdata[p0_, e0_, y_, tf0_] := Table[{Evaluate[hplus[x] /. sol[p0, e0, y]],
   Evaluate[hcross[x] /. sol[p0, e0, y]], x}, {x, 0, tf0}]
```

In[12]:=
```
hplusdata[50, 0.7, 10^-7, 100];
```

Plotters

In[•]:=
```
?wfsdsgwplot
```

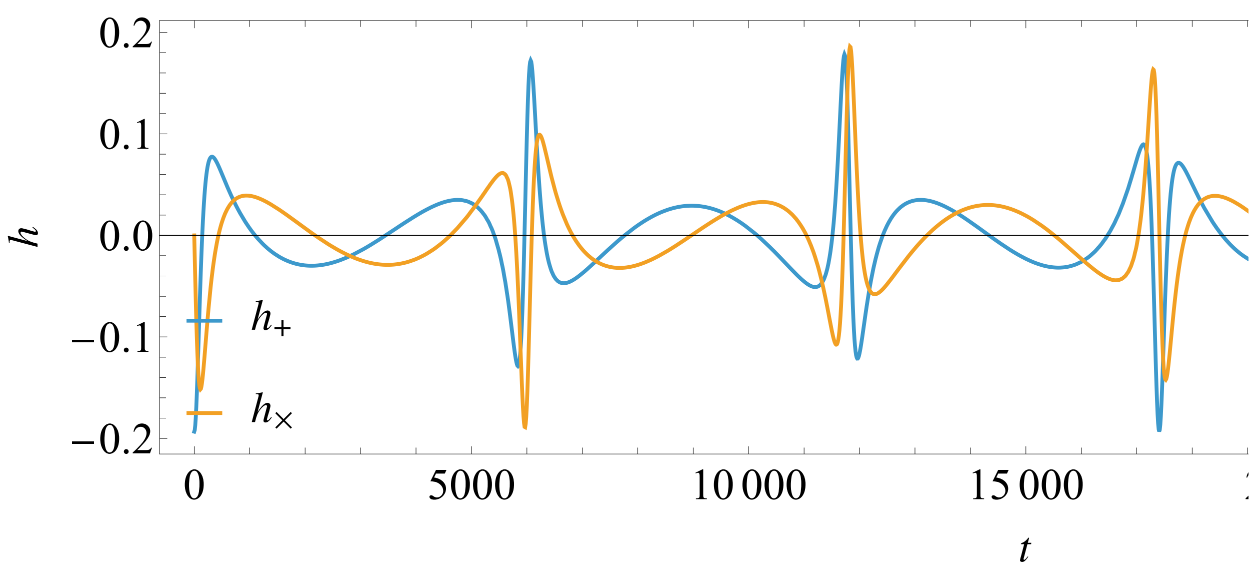
Out[•]=

Symbol

wfsdsgwplot[p0_,e0_,y0_,tf0_] plots the plus and cross polarization

In[•]:=
```
wfsdsgwplot[p0_, e0_, y0_, tf0_] := Module[{tf, p, e, y},
  p = p0;
e = p0;
y = y0;
tf = tf0;
  Plot[{Evaluate[hplus[t] /. sol[p0, e0, 0]], Evaluate[hcross[t] /. sol[p0, e0, 0]]},
   {t, 0, tf0}, Frame → True, AspectRatio → 1/4, ImageSize → 1000,
   PlotRange → {Automatic, Full}, LabelStyle → style,
   Frame → True, FrameLabel → {MaTeX["t"], MaTeX["h"]},
   PlotLegends → Placed[{MaTeX["h_+"], MaTeX["h_\\times"]}, {Left, Bottom}]]]
wfsdsgwplot[50, 0.7, 10^-7, 30000]
```

Out[•]=



In[14]:=
```
?wfsdsgwplusplot
```

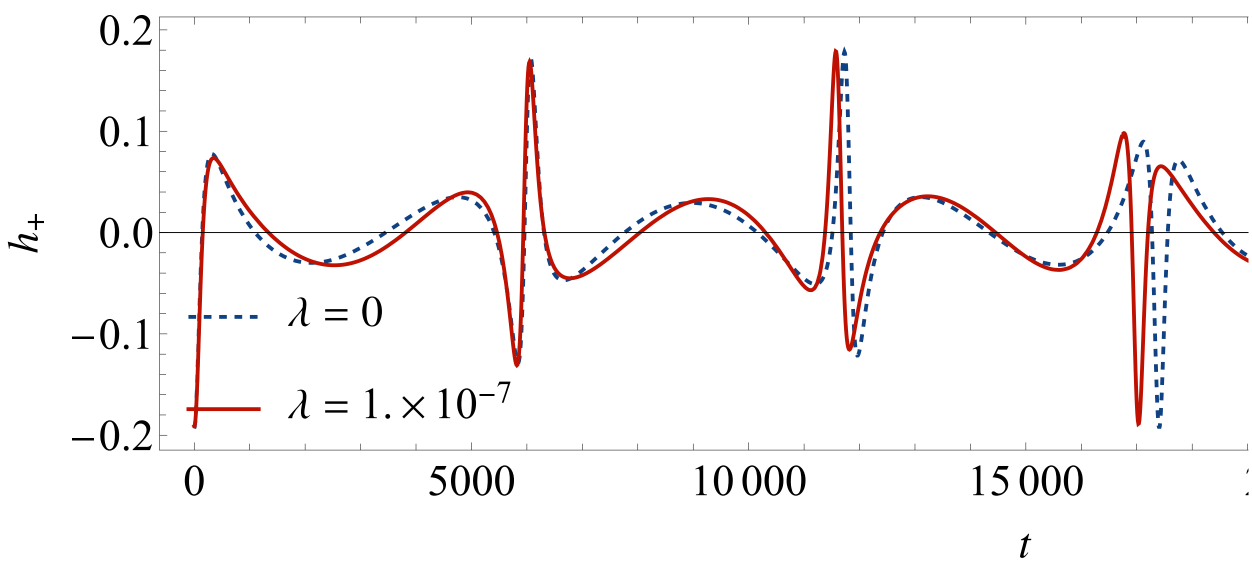
Out[14]=

Symbol

wfsdsgwplusplot[p0_,e0_,y0_,tf0_] plots the plus polarization against a reference 0pN waveform

In[20]:=
```
wfsdsgwplusplot[p0_, e0_, y0_, tf0_] := Module[{tf, p, e, y},
  p = p0;
e = p0;
y = y0;
tf = tf0;
  Plot[{Evaluate[hplus[t] /. sol[p0, e0, 0]], Evaluate[hplusSdS[y, t] /. sol[p0, e0, y]]},
   {t, 0, tf0}, Frame → True, AspectRatio → 1/4, ImageSize → 1000,
   PlotRange → {Automatic, Full}, LabelStyle → style, Frame → True,
   FrameLabel → {MaTeX["t"], MaTeX["h_+"]}, PlotStyle → {{Dashed, Thick,
      RGBColor[0.06274509803921569, 0.25882352941176473`, 0.5137254901960784]},
     {Thick, RGBColor[0.7450980392156863, 0.06666666666666667, 0.00392156862745098]}},
   PlotLegends → Placed[{MaTeX["\\lambda=0"],
      MaTeX["\\lambda="] × TraditionalForm[y0 // N]}, {Left, Bottom}]]]
wfsdsgwplus = wfsdsgwplusplot[50, 0.7, 10^-7, 30000]
```

Out[21]=



In[25]:=
```
exportPath = FileNameJoin[{plotsFolder, "plus_50_7_-7.pdf"}];
Export[exportPath, wfsdsgwplus];
```

In[15]:=
```
?wfsdsgwcrossplot
```

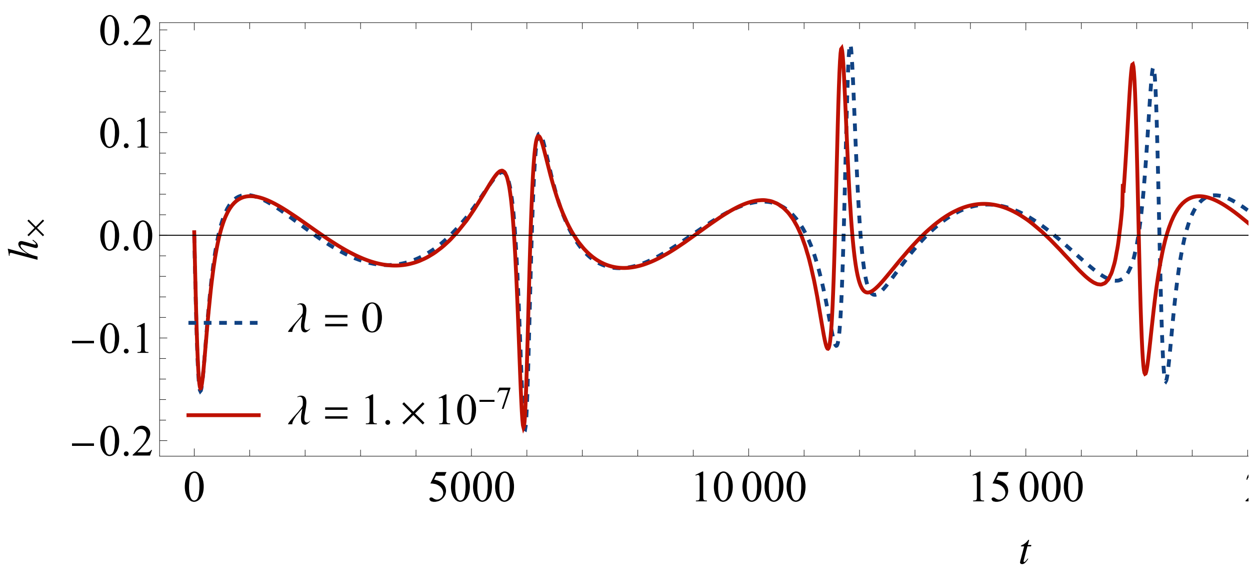
Out[15]=

Symbol

wfsdsgwcrossplot[p0_,e0_,y0_,tf0_] plots the cross polarization against a reference 0pN waveform

In[27]:=
```
wfsdsgwcrossplot[p0_, e0_, y0_, tf0_] := Module[{tf, p, e, y},
  p = p0;
e = p0;
y = y0;
tf = tf0;
  Plot[{Evaluate[hcross[t] /. sol[p0, e0, 0]],
    Evaluate[hcrossSdS[y, t] /. sol[p0, e0, y]]}, {t, 0, tf0}, Frame → True, AspectRatio → 1/4,
   ImageSize → 1000, PlotRange → {Automatic, Full}, LabelStyle → style, Frame → True,
   FrameLabel → {MaTeX["t"], MaTeX["h_\\times"]}, PlotStyle → {{Dashed, Thick,
      RGBColor[0.06274509803921569, 0.25882352941176473`, 0.5137254901960784]},
     {Thick, RGBColor[0.7450980392156863, 0.06666666666666667, 0.00392156862745098]}},
   PlotLegends → Placed[{MaTeX["\\lambda=0"],
      MaTeX["\\lambda="] × TraditionalForm[y0 // N]}, {Left, Bottom}]]]
wfsdsgwcross = wfsdsgwcrossplot[50, 0.7, 10^-7, 30000]
```

Out[28]=



In[29]:=
```
exportPath = FileNameJoin[{plotsFolder, "cross_50_7_-7.pdf"}];
Export[exportPath, wfsdsgwplus];
```

Dephasing

The dephasing of the orbit and the waveform may simply be obtained from the phase of the orbit subtracted from a reference phase

cumulative dephasing = $|\ \Phi_{\mathrm{SdS}}(t) - \Phi_{\mathrm{pN}}(t)\ |$

In[33]:=
```
Plot[
  Abs[Evaluate[(f[x] + δ[x])/π /. sol[50, 0.7, 0]] - Evaluate[(f[x] + δ[x])/π /. sol[50, 0.7, 10^-7]]],
  {x, 0, 30000}, Frame → True, LabelStyle → style, Frame → True,
  FrameLabel → {MaTeX["t"], MaTeX["\\text{Dephasing (in radians)}"]}, ImageSize → Large]
```

Out[33]=